

Tutorial Problem Set IX

MATA30 – TUT0003

OLLE RENÉRIUS REHNQUIST

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This tutorial was somewhat different from our usual tutorials due to the upcoming midterm. We solved the problems from the practise midterm and discussed optimization. I have added some problems about some of the main theorems in the course, as this will be relevant to the test. Good luck!

1 Central theorems

1.1 Extreme value theorem

- (a) State the extreme value theorem.
- (b) Does it still hold if the function is not continuous?
- (c) Does it still hold if the domain is an open interval?
- (d) Is the extrema necessarily strict?

1.2 Mean value theorem

- (a) State Rolle's theorem.
- (b) State and prove the mean value theorem from Rolle's theorem.
- (c) Is continuity at the endpoints of the interval necessary for the mean value theorem?

1.3 First and second derivative test

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function and $a \in \mathbb{R}$. Which of the following statements are true?

- (a) If f obtains a maximum at a , then $f'(a) = 0$.
- (b) If $f'(a) = 0$, then f obtains a global maximum or minimum at a .
- (c) If $f'(a) = 0$, then f obtains a local maximum or minimum at a .
- (d) If $f'(a) = 0$ and $f''(a) > 0$, then f obtains a local minimum at a .
- (e) If $f'(a) = 0$ and $f''(a) = 0$, then f does *not* obtain a local maximum or minimum at a .